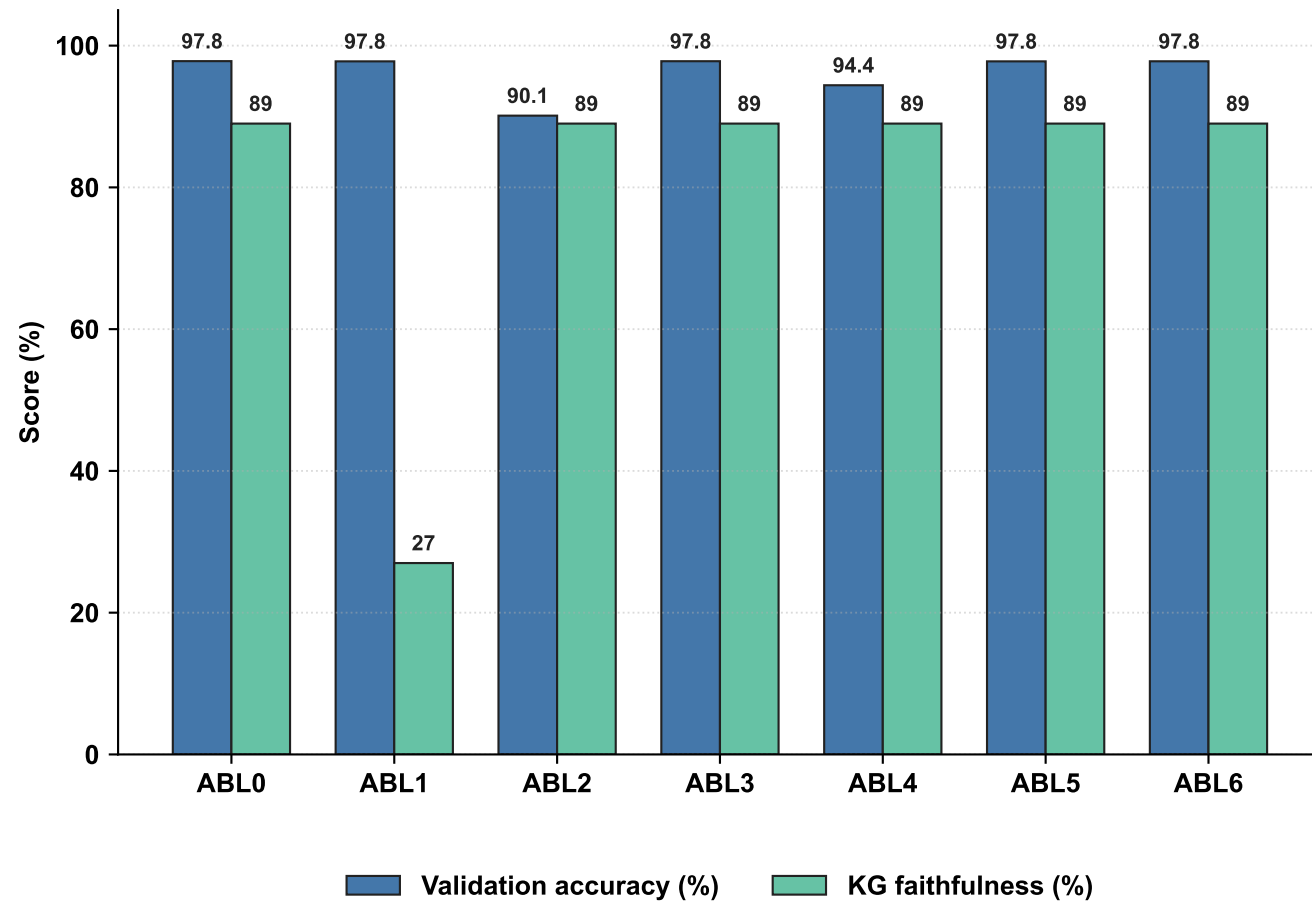
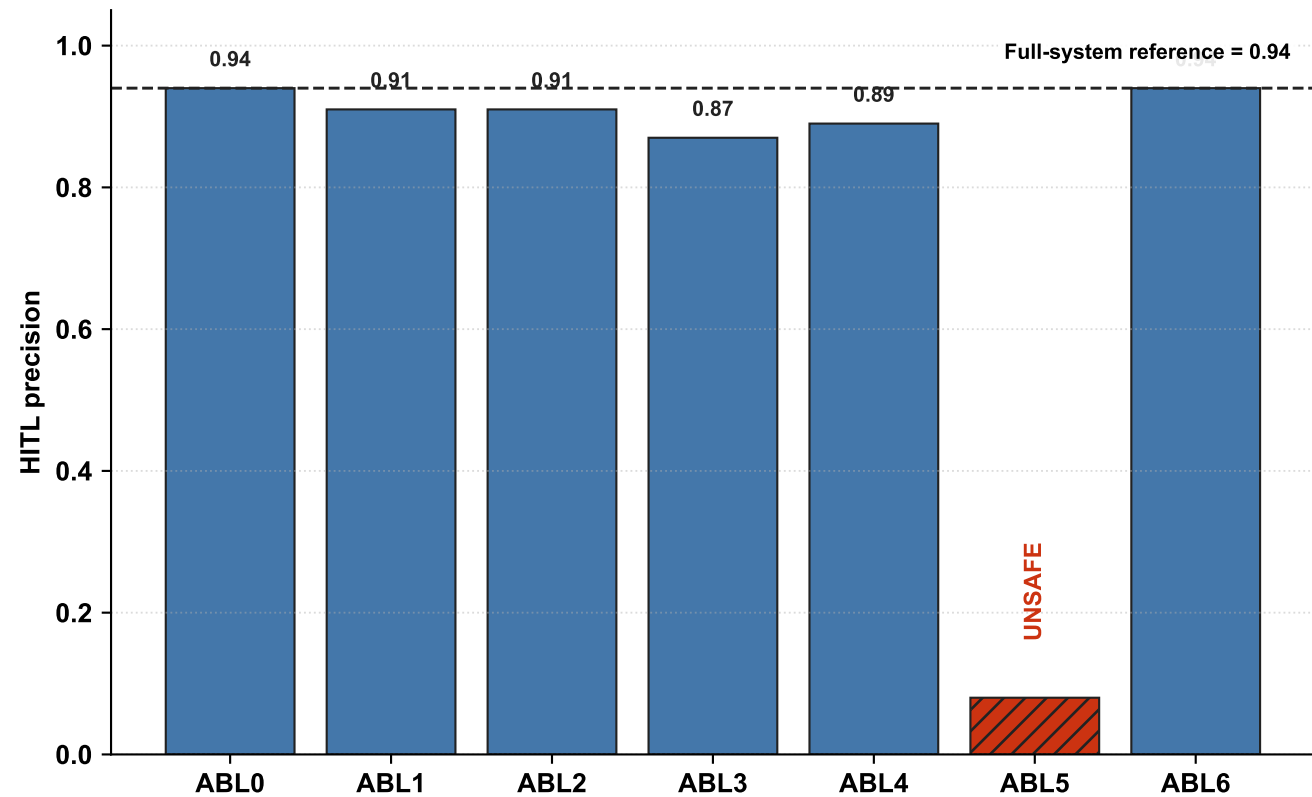


(a) Predictive performance and explanation faithfulness



(b) HITL routing precision under component removal



ABL0 = full system; ABL1 = no KG; ABL2 = no speech; ABL3 = no DT; ABL4 = no cross-attention; ABL5 = no HITL; ABL6 = no privacy gate.

Ablation results distinguish predictive performance, explanation faithfulness, and safety-routing contribution.